

Event vs full-path vs cycle replanning for three planners

Rebuilding more often costs more planning time without finishing more runs (every arm finishes 197–200 of 200): at 20 ms checks Visibility Graph rebuilds 26.5 (event), 53.6 (full-path) and 119.7 (cycle) times per run. No rebuild took over 100 ms, so no run failed on computing time; the runs that failed had no route for 1 s (obstacles blocked every path): event 1, full-path 9, cycle 6. Event (ours) rebuilds when the next segment is blocked; full-path also when any later segment is (Costa and Tonidandel 2024); cycle rebuilds at every check. Brackets show the change from event. In each group of three, ▲ blue (bold) marks the best value and ▼ orange the worst; planning time and time per rebuild are marked only when they differ by more than 20 %, the drift between batches. The rebuild-over-100 ms column depends on the machine; the no-route column does not.

	Runs finished (of 200)	Finished without contact (of 200)	Rebuilds per run	Rebuilds from blocked later segment (/run)	Planning time per run (ms)	Time per rebuild, median (ms)	Failed runs: rebuild over 100 ms	Failed runs: no route for 1 s	Time a route is kept (ms)
Voronoi									
Event (ours) @ 20 ms	200	95 ▼	7.1	0.2	102	8.7	0	0	606 ▲
Full-path @ 20 ms	200	95 ▼	7.1 (−0.1) ▲	0.1 (−0.1)	101 ▲	8.2 (−0.5)	0	0	603 (−3)
Cycle @ 20 ms	200	102 (+7) ▲	81.2 (+74.0) ▼	–	677 (+576) ▼	8.7	0	0	140 (−466) ▼
Event (ours) @ 100 ms	200	74 ▼	4.1	0.1	42	8.9	0	0	838
Full-path @ 100 ms	200	74 ▼	4.1 ▲	0.0	40 (−2) ▲	8.2 (−0.7)	0	0	839 ▲
Cycle @ 100 ms	200	85 (+11) ▲	15.4 (+11.3) ▼	–	130 (+88) ▼	8.8 (−0.1)	0	0	497 (−341) ▼
PRM									
Event (ours) @ 20 ms	200 ▲	145 ▼	7.7 ▲	0.0 ▲	182 ▲	23.5	0	0 ▲	553 ▲
Full-path @ 20 ms	197 (−3) ▼	147 (+2)	12.8 (+5.1)	4.7 (+4.6) ▼	272 (+90)	22.5 (−1.0)	0	3 (+3) ▼	432 (−120)
Cycle @ 20 ms	197 (−3) ▼	154 (+9) ▲	78.6 (+70.8) ▼	–	1804 (+1622) ▼	23.3 (−0.2)	0	3 (+3) ▼	143 (−409) ▼
Event (ours) @ 100 ms	199 ▲	92 ▼	4.6 ▲	0.0 ▲	96 ▲	23.2	0	1 ▲	781 ▲
Full-path @ 100 ms	198 (−1) ▼	95 (+3)	7.1 (+2.5)	2.8 (+2.8) ▼	137 (+41)	22.3 (−0.9)	0	2 (+1) ▼	681 (−100)
Cycle @ 100 ms	198 (−1) ▼	98 (+6) ▲	15.2 (+10.6) ▼	–	349 (+253) ▼	23.4 (+0.2)	0	2 (+1) ▼	484 (−296) ▼
Visibility Graph									
Event (ours) @ 20 ms	200 ▲	139 ▲	26.5 ▲	0.0 ▲	541 ▲	21.0	0	0 ▲	321 ▲
Full-path @ 20 ms	198 (−2) ▼	136 (−3)	53.6 (+27.1)	29.0 (+28.9) ▼	952 (+411)	18.1 (−2.9)	0	2 (+2) ▼	191 (−130)
Cycle @ 20 ms	200 ▲	135 (−4) ▼	119.7 (+93.2) ▼	–	2339 (+1797) ▼	20.4 (−0.7)	0	0 ▲	101 (−220) ▼
Event (ours) @ 100 ms	200 ▲	82 ▼	8.4 ▲	0.0 ▲	165 ▲	20.6	0	0 ▲	658 ▲
Full-path @ 100 ms	198 (−2) ▼	84 (+2) ▲	14.6 (+6.2)	6.3 (+6.3) ▼	259 (+95)	18.6 (−2.0)	0	2 (+2) ▼	506 (−152)
Cycle @ 100 ms	199 (−1)	82 ▼	21.4 (+13.0) ▼	–	415 (+250) ▼	20.1 (−0.5)	0	1 (+1)	410 (−248) ▼